Bybit L2 Order-Flow Feasibility Demonstration

Client guide: installation, running the demo, what is inside

https://github.com/ibrahimAltaf/bybit-l2-orderflow-demo

Prepared 2026-11-02 | Real Bybit historical data | Python 3.12

At a glance

1,265,452	3 / 3	360,503	0
real Bybit L2 deltas parsed (3 days; u always previous u + 1)	rebuilt books identical to end-of-file snapshots	events replayed (ORDIUSDT, full UTC day)	mismatches in independent recheck of 6,086 transitions

This repository shows, on real data, that Bybit's free official archive provides incremental Level-2 order-book data (snapshot + deltas with update IDs) and trades for USDT perpetuals, including smaller and delisted contracts; that the order book can be rebuilt exactly; and that a deterministic, look-ahead-free event replay can drive an auditable LONG / SHORT **WATCH -> CANDIDATE -> CONFIRMED** signal state machine.

Scope. Signal research only. No execution, risk or exit logic is included, and your existing engine (TP1 / TP2 / ATR-Chandelier trail / optional break-even) is not touched. Thresholds are a-priori and untuned. **No profitability is claimed.**

1. Requirements

Item	Needed
Operating system	Windows 10/11, macOS or Linux
Python	3.12 (exactly 3.12.x; *python.org* installer is fine)
Tools	git
Network	Internet on first run (~70 MB is downloaded from Bybit and Tardis.dev)
Accounts / API keys	None. No environment variables are required.
Disk / memory	~150 MB disk; under 100 MB RAM during the replay (78.5 MB peak measured)

2. Installation

Windows (PowerShell or Command Prompt)

```
git clone https://github.com/ibrahimAltaf/bybit-l2-orderflow-demo.git
cd bybit-l2-orderflow-demo
py -3.12 -m venv .venv
.venv\Scripts\activate
pip install -e ".[dev]"
```

macOS / Linux

```
git clone https://github.com/ibrahimAltaf/bybit-l2-orderflow-demo.git
cd bybit-l2-orderflow-demo
python3.12 -m venv .venv
source .venv/bin/activate
pip install -e ".[dev]"
```

> Tip: if PowerShell blocks **activate**, run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` once, or use Command Prompt.

3. Run the demo (one command)

```
python scripts/run_demo.py
```

The first run downloads the data, verifies it, replays one full trading day twice and re-checks the results with an independent script. It takes about 6-12 minutes depending on the machine and network. It prints ten numbered stages, each marked PASS or FAIL, and ends with one line:

```
DEMO RESULT: PASS
```

Stage	What is checked
1 Environment	Python version, package version, configuration hash
2 Historical data	Download (or verified cache) of 8 real files; every record parsed
3 Hash validation	Every raw file's SHA-256 equals the recorded manifest
4 L2 reconstruction	Rebuilt book equals the end-of-file snapshot, level for level
5 Continuity	Update-ID sequence, duplicates, gaps, crossed books, timestamps
6 Trade flow	Trade side vs. book; cross-check with a second provider (Tardis.dev)

Stage	What is checked
7 Event replay	Full ORDIUSDT day, exchange-time order, strict schema validation
8 State machine	Every transition logged; CONFIRMED signals listed
9 Rejections	Rejected setups by stage, and every rejection reason
10 Reproducibility	Byte-identical second run; independent no-look-ahead recheck

4. What is inside the repository

Path	Contents
scripts/run_demo.py	The one-command demonstration (stages 1-10, PASS/FAIL)
scripts/independent_recompute.py	Independent check using only the Python standard library
scripts/benchmark_replay.py	Runtime, throughput and peak-memory measurement
config/demo.yaml	Datasets and every detector threshold, with explanations
src/bybit_demo/data/	Download with SHA-256 manifest, validation, parsing, verification
src/bybit_demo/book/	Local L2 order book, continuity / gap / resync rules, liquidity metrics
src/bybit_demo/flow/	CVD and rolling order-flow windows (event-driven, no candles)
src/bybit_demo/features/	Book pressure, absorption, liquidity, volatility, extension
src/bybit_demo/detector/	LONG and mirrored SHORT WATCH / CANDIDATE / CONFIRMED machine
src/bybit_demo/replay/	Deterministic chronological replay engine
src/bybit_demo/audit/	Audit trail writers (JSONL, CSV) and evidence manifest
tests/	151 automated tests incl. adversarial cases; synthetic fixtures (labelled)
docs/	Reports: data feasibility, technical results, architecture, limitations, review
data/evidence/	Evidence written by the demo (see section 5)

5. Evidence files produced by the demo

File	What it proves
download_manifest.json	URL, size, SHA-256 and retrieval time of every raw file
dataset_verification.json	Full-file checks per dataset: schema, update IDs, checkpoints, trade side
provider_cross_check.json	Bybit archive vs. Tardis.dev: trades by ID, top of book per minute
demo_events.jsonl	Every state transition with all features and thresholds (audit trail)
signals.csv	One compact row per transition
continuity_report.json	Continuity rules applied and every issue found
replay_summary.json	Counts: events, transitions, signals, rejections
determinism_check.json	SHA-256 of two independent runs' outputs (identical)
independent_recompute.json	Standard-library recheck of logged features from raw files
evidence_manifest.json	SHA-256 of every evidence file

demo_events.jsonl and signals.csv contain exchange prices and are regenerated locally by the demo rather than published in the repository.

6. Results on real data

Verified Bybit datasets (every record parsed)

Symbol	Day	Depth	Deltas	u gaps	Snapshot match	Trades	Taker-side fit
ORDIUSDT	2026-10-01	ob200	339,154	0	1/1	21,347	95.8%
10000LADYSUSDT delisted	2025-10-01	ob200	553,441	0	1/1	48,184	91.5%
10000LADYSUSDT delisted	2025-06-01	ob500	372,857	0	1/1	22,605	93.7%

Second provider: Tardis.dev holds the identical 21,347 ORDIUSDT trades (0 differences in price, size or side); top of book identical at 1,383 of 1,439 minute marks.

Signal state machine (ORDIUSDT, 2026-10-01)

6,086	7	2,666	IDENTICAL
logged transitions	CONFIRMED (1 LONG, 6 SHORT)	rejected setups (each with reasons)	two independent runs

Signal time (UTC)	Direction	Update ID
2026-10-01 03:44:39.214000	LONG	26831968
2026-10-01 07:36:03.408000	SHORT	26877172
2026-10-01 11:52:58.590000	SHORT	26942560
2026-10-01 14:50:45.408000	SHORT	26996396
2026-10-01 15:20:52.356000	SHORT	27007155
2026-10-01 16:51:36.884000	SHORT	27034938
2026-10-01 16:56:41.705000	SHORT	27036367

These are engineering outputs of untuned rules on one day of one contract, not hit rates or returns. No forward price outcome was computed.

7. Useful individual commands

```
python scripts/run_demo.py --skip-verify          # reuse earlier verification
python -m bybit_demo.cli verify-data             # download + verify datasets
python -m bybit_demo.cli replay                  # replay the primary day
python -m bybit_demo.cli replay --out out/ --start 2026-10-01T03:00:00Z \
                                --end 2026-10-01T04:00:00Z
python scripts/independent_recompute.py          # standard-library recheck
python -m pytest                                 # automated tests
python scripts/benchmark_replay.py               # performance
```

8. Troubleshooting

Symptom	Cause and fix
'python' is not 3.12	Use `py -3.12` (Windows) or `python3.12` to create the venv.
Network error during stage 2	Downloads retry automatically; without a connection the demo stops with an explicit message. Re-run when online.
IntegrityError	A provider file changed since the recorded hash; the demo refuses to use it so the evidence stays valid.
CorruptFileError	A damaged or partial archive was detected and rejected. Re-run: a cached file that fails its hash or integrity check is downloaded again automatically.
Tests show 'skipped'	Real-data tests skip until the demo has downloaded the data.
Run from another folder	Run commands from the repository root (the scripts handle this).

9. Limitations to keep in mind

- One symbol-day is replayed through the detector; nothing is generalised.
- Thresholds are a-priori engineering values, deliberately not tuned.
- No profitability, backtest or out-of-sample result is claimed.
- The ob200 book is published in 100 ms batches, so execution vs. cancellation attribution is approximate.
- Update-ID +1 continuity was observed on the archive, not documented by Bybit.
- Open interest, funding and liquidation data are planned, not yet included.

Full list: **docs/LIMITATIONS.md**; self-review: **docs/ADVERSARIAL_REVIEW.md**.

10. Next steps for the awarded project

Area	Planned work
Coverage	Multi-symbol, multi-day archive scans and universe selection
Live data	WebSocket ingestion with gap detection and resync
More inputs	Open interest, funding, liquidations, regime features
Validation	Threshold calibration and out-of-sample testing across many symbols/days
Integration	Signals into your existing bot, without changing its execution, risk or exits